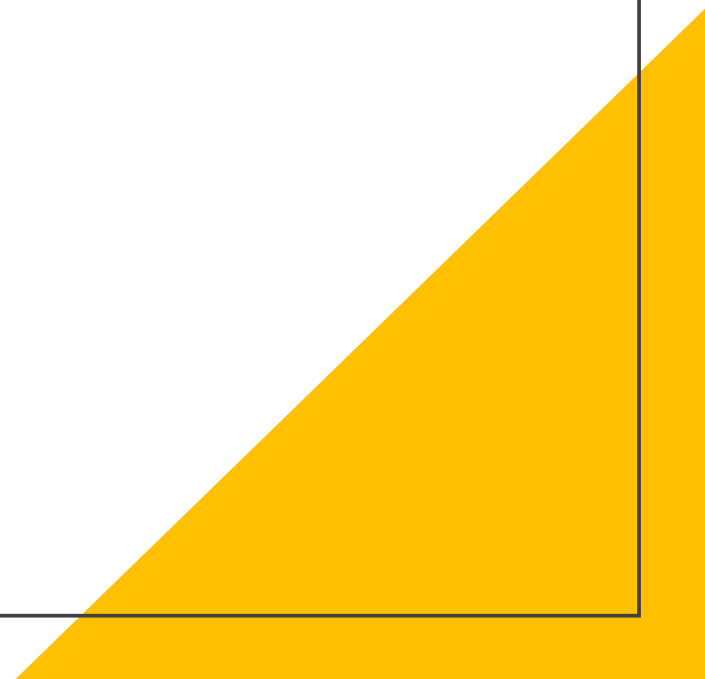


資料排序



資料排序

有時候後端來的資料沒有依照我們希望的順序進行排序，這時候我們就會需要在顯示資料前先將資料排序後再讓畫面顯示，排序通常會需要一個值可能是數字(id)或者文字，數字的話當然就是由小到大或者由大到小，文字的話就是會去判斷第一個文字由A > Z或者Z > A，中文無法除非要自己寫額外的方法去做排序跟判斷。

資料排序

我們看到下面的資料，如果沒有排序直接使用頁面的@for將其顯示在畫面上時，第一筆資料就會是position 10的那筆資料。

```
let ELEMENT_DATA: PeriodicElement[] = [  
  {position: 10, name: 'Neon', weight: 20.1797, symbol: 'Ne'},  
  {position: 1, name: 'Hydrogen', weight: 1.0079, symbol: 'H'},  
  {position: 2, name: 'Helium', weight: 4.0026, symbol: 'He'},  
  {position: 3, name: 'Lithium', weight: 6.941, symbol: 'Li'},  
  {position: 4, name: 'Beryllium', weight: 9.0122, symbol: 'Be'},  
  {position: 5, name: 'Boron', weight: 10.811, symbol: 'B'},  
  {position: 6, name: 'Carbon', weight: 12.0107, symbol: 'C'},  
  {position: 7, name: 'Nitrogen', weight: 14.0067, symbol: 'N'},  
  {position: 8, name: 'Oxygen', weight: 15.9994, symbol: 'O'},  
  {position: 9, name: 'Fluorine', weight: 18.9984, symbol: 'F'},  
];
```

資料排序

這時候我們就要使用到sort()這個方法注意這個方法要陣列才能使用，看到這邊的寫法這邊是使用position去做為排序的判斷，sort會一直將第一筆跟第二筆資料去做比較(持續下去直到比對到最後一筆)。

```
this.dataSource.paginator = this.paginator;  
this.dataSource.data.sort(function (a, b) {  
  if (a.position < b.position) {  
    return -1;  
  }  
  if (a.position > b.position) {  
    return 1;  
  }  
  return 0;  
});
```

資料排序

這邊看到如果我的第一筆(a)的position小於(b)的position就會回傳-1也就是將其排序往後遞延反之則往前。

```
this.dataSource.paginator = this.paginator;  
this.dataSource.data.sort(function (a, b) {  
  if (a.position < b.position) {  
    return -1;  
  }  
  if (a.position > b.position) {  
    return 1;  
  }  
  return 0;  
});
```

資料排序

所以最後你就會得到這樣的資料。

```
▶ 0: {position: 1, name: 'Hydrogen', weight: 1.0079, symbol: 'H'}  
▶ 1: {position: 2, name: 'Helium', weight: 4.0026, symbol: 'He'}  
▶ 2: {position: 3, name: 'Lithium', weight: 6.941, symbol: 'Li'}  
▶ 3: {position: 4, name: 'Beryllium', weight: 9.0122, symbol: 'Be'}  
▶ 4: {position: 5, name: 'Boron', weight: 10.811, symbol: 'B'}  
▶ 5: {position: 6, name: 'Carbon', weight: 12.0107, symbol: 'C'}  
▶ 6: {position: 7, name: 'Nitrogen', weight: 14.0067, symbol: 'N'}  
▶ 7: {position: 8, name: 'Oxygen', weight: 15.9994, symbol: 'O'}  
▶ 8: {position: 9, name: 'Fluorine', weight: 18.9984, symbol: 'F'}  
▶ 9: {position: 10, name: 'Neon', weight: 20.1797, symbol: 'Ne'}
```

資料排序

當你想要比對的是另一個欄位就直接將他的欄位修改。

```
this.dataSource.data.sort(function (a, b) {  
  if (a.name < b.name) {  
    return -1;  
  }  
  if (a.name > b.name) {  
    return 1;  
  }  
  return 0;  
});
```

練習

將下圖的資料用重量排序並且打印出來。

```
export class LevelComponent {  
  ELEMENT_DATA = [  
    {position: 10, name: 'Neon', weight: 20.1797, symbol: 'Ne'},  
    {position: 1, name: 'Hydrogen', weight: 1.0079, symbol: 'H'},  
    {position: 2, name: 'Helium', weight: 4.0026, symbol: 'He'},  
    {position: 3, name: 'Lithium', weight: 6.941, symbol: 'Li'},  
    {position: 4, name: 'Beryllium', weight: 9.0122, symbol: 'Be'},  
    {position: 5, name: 'Boron', weight: 10.811, symbol: 'B'},  
    {position: 6, name: 'Carbon', weight: 12.0107, symbol: 'C'},  
    {position: 7, name: 'Nitrogen', weight: 14.0067, symbol: 'N'},  
    {position: 8, name: 'Oxygen', weight: 15.9994, symbol: 'O'},  
    {position: 9, name: 'Fluorine', weight: 18.9984, symbol: 'F'},  
  ]  
}
```


结束

